Practice Test 01

Q.1

Given $\vec{F}(x, y) = (y, -x)$:

(a) Sketch or describe the general direction of this field.

(b) Find its magnitude at the point $(1, 0)$.

(c) Is this vector field conservative? Justify.

Q.2

Evaluate the line integral:

$$\int_C (x^2 + y^2)\, ds$$

where C is the upper semicircle $x = \cos t,\ y = \sin t,\ 0 \le t \le \pi.$

Q.3

Compute the work done by the force field

$$\vec{F}(x, y) = (y, x)$$

along the path from $(0, 0)$ to $(1, 1)$ along the line $y = x$.

Q.4

Find the circulation of $\vec{F}(x, y) = (-y, x)$ around the unit circle $x^2 + y^2 = 1$, counterclockwise.

Q.5

Find the **outward flux** of $\vec{F}(x, y) = (x, y)$ across the boundary of the unit circle $x^2 + y^2 = 1$.

(Hint: The outward normal vector to a circle is $\hat{n} = (x, y)$ on the boundary.)

Q.6

Let $f(x, y) = x^2 y + y^3$.

Compute ∇f and then evaluate

$$\int_C \nabla f \cdot d\vec{r}$$

where C is any smooth path from $(1, 0)$ to $(2, 2)$.

Q.7

Determine whether the field

$$\vec{F}(x, y) = (2xy, x^2 + 2y)$$

is conservative. If it is, find its potential function $f(x, y)$.

Q.8

Let $\vec{F}(x, y) = (x^2 - y,\, x + y^2)$.

(a) Determine if $\vec{F}$ is conservative.

(b) If yes, find $f(x, y)$ such that $\vec{F} = \nabla f$.

(c) Use the FTLI to compute $\int_C \vec{F} \cdot d\vec{r}$ for a path from $(0, 0)$ to $(1, 1)$.